

VLSI DESIGN

Engineering the Structural Logic of Integrated Circuits & SoC Development

A project-first certification designed to master RTL Logic Design, SystemVerilog Verification, and the structural engineering of Physical Design. Master the silicon development lifecycle from gate-level logic to GDSII tape-out.



THE EDLANX METHODOLOGY

OUR VLSI DESIGN PHILOSOPHY

At Edlanx, we don't just teach "chip design"; we cultivate a Silicon Architect's mindset. Our program is a deep dive into the end-to-end VLSI design flow. We believe in Deterministic Performance — moving beyond theoretical gate logic to build high-performance, low-power integrated circuits that meet the rigorous timing and area constraints of the 2026 semiconductor market.

THE FOUR PILLARS OF MASTERY

-  **Architectural Logic** Master the structural blueprinting of systems using HDLs like Verilog and VHDL.
-  **Verification Governance** Learn the logic of UVM and SystemVerilog to ensure first-pass silicon success.
-  **Physical Architecture** Master the structural flow of Floorplanning, Placement, and Routing (PnR).
-  **Timing & Power Strategy** Bring your designs to life by mastering Static Timing Analysis (STA) and Power-Performance-Area (PPA) optimization.

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WHY CHOOSE EDLANX



01 Industry-Ready Curriculum

We move beyond basic gate logic to bridge the gap between academic electronics and foundry-ready engineering. Our VLSI syllabus is a deep immersion into RTL Design, UVM Verification, and Physical Design — the exact technical pillars used by global semiconductor giants and fabless design houses.

02 Real-Time Corporate Exposure

Step into a professional chip design environment at our Bangalore Operational Center. At Edlanx, you won't just run simulations; you will be treated as a Design Engineer, working within real-world constraints to optimize Power-Performance-Area (PPA), solve timing violations, and manage complex clock trees.

03 Mentorship & Continuous Assessment

Your growth is guided by senior Silicon Architects and Verification Leads through structured mentoring and performance evaluations. We focus on developing your ability to write synthesizable code, architect robust testbenches, and perform Static Timing Analysis (STA) with the mathematical precision required for modern SoC development.

04 Skill Mastery & Performance Incentives

Master industry-standard EDA (Electronic Design Automation) tools while learning essential professional competencies. We believe in experiential learning and reward innovative breakthroughs in low-power design or verification coverage with performance-based stipends, ensuring you remain focused on industrial excellence.

05 Global Career Readiness

We enhance your employability by focusing on the complete Silicon Development Lifecycle, from initial architectural specs to final GDSII tape-out. Our career guidance includes mock "Technical Design Reviews" tailored for top-tier roles in RTL design, physical design, and pre-silicon verification.

06 Collaborative Engineering Community

Join a supportive culture built on transparency, technical integrity, and disciplined ethics. You will learn to lead a design "sprint," collaborate across front-end and back-end domains, and grow your network within an elite community of VLSI professionals and hardware specialists.

07 Verified Professional Certifications

At Edlanx, we don't just issue certificates for attendance; we award them based on verified learning, real-world performance, and practical outcomes. Our credentials are proof of your industry-ready skills and are recognized by institutions and recruiters worldwide.

We provide four distinct levels of certification:

- Internship Completion Certificate: Validates your exposure to professional work culture, verified internship hours, and task-based mentorship.
 - Project Completion Certificate: Proof of your hands-on implementation and problem-solving skills in real-world industry projects.
 - Training Completion Certificate: Issued upon mastering our industry-aligned curriculum and passing skill-based assessments.
 - Certificate of Excellence: A prestigious award reserved for top-performing candidates who demonstrate an advanced understanding.
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Why This Journey Matters

-  **Portfolio-Ready** Every synthesizable RTL module, UVM verification environment, and physical GDSII layout you architect at Edlanx serves as a high-fidelity demonstration of your readiness for the global semiconductor stage.
-  **Global Network** Gain exclusive access to the Edlanx alumni community and industry mentors leading design teams at top-tier fabless companies, foundry giants, and cutting-edge hardware startups.
-  **Career Excellence** We move beyond basic gate logic to provide the technical resume engineering and system-integration logic needed to lead the future of "Intelligent Silicon" and the global surge in specialized AI hardware.

This course is a living ecosystem. We constantly update our curriculum to reflect the latest shifts in SystemVerilog standards, RISC-V extensions, and nanometer-scale Physical Design challenges. Your success is our shared mission.

DIRECT MENTORSHIP



Engineering a reliable integrated circuit requires absolute mathematical and timing precision. Whether you are debugging a complex race condition in a multi-clock domain, optimizing the power-grid for IR drop, or closing timing on a critical path, our senior Silicon Architects are available to provide the direct, technical feedback needed to ensure your designs are foundry-ready.

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